Lin Ding Shan

AgriTech Computer Vision | Edge AI Deployment

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Kuala Lumpur, Malaysia · Available July 2027 · Relocating to New Zealand or Australia

PROFESSIONAL SUMMARY

Computer-vision engineer who takes detection models from orchard photos to working tools for growers. Built SnapDurian, a durian disease and pest scouting app that runs in a phone browser and started as a handheld Raspberry Pi 5 device, and, as an intern, a disease-monitoring node for a commercial greenhouse. Published a peer-reviewed paper on the device in *Smart Agricultural Technology* (Elsevier). Final-year Data Science student at UCSI University, graduating May 2027.

TECHNICAL SKILLS

Programming: Python, SQL, Bash; Git / GitHub
Computer vision: PyTorch, Ultralytics YOLO (v8, 11), RT-DETR, Faster R-CNN, MobileNet, EfficientNet, OpenCV
Deployment: ONNX Runtime, LiteRT / TFLite, INT8 and weight-only quantisation; Raspberry Pi 5, Jetson Orin Nano (CUDA), Linux, libcamera, V4L2
Web & backend: FastAPI, SQLite, Docker Compose, Caddy (HTTPS), PWA (service worker, offline queue), Telegram Bot API
Performance engineering: Latency, power and thermal telemetry; battery / UPS telemetry over I2C; hardware performance counters; duty cycling; multi-model scheduling
Data: Field data collection, annotation protocols, dataset merging

EXPERIENCE

Computer Vision Engineer | SnapDurian (snapdurian.com), independent agri-tech venture, Malaysia *Jul 2025 – Present*

- Built SnapDurian as a phone web app (PWA): a grower photographs a leaf or trunk and sees the likely disease or pest among 11 categories marked on the photo, in English, Chinese or Malay; photos taken offline sync when signal returns. **Onboarded 4 pilot growers** in Sabah, Negeri Sembilan and Johor.
- Photographed a farm-labelled dataset at 12 orchard sites in Peninsular Malaysia and Sabah from Dec 2025; agreed the lesion-level annotation standard with a crop-protection practitioner, managed outsourced annotation and reviewed every image. Merged the per-category annotation exports into one product set (~2,000 images, ~23,000 lesion boxes).
- Deployed the backend (FastAPI, ONNX Runtime, Docker, Caddy HTTPS) on a CPU cloud server, with invite codes and daily limits for the pilot and model swaps without restart.
- Added a tiled second pass for small lesions in distant shots; one-tap grower feedback becomes pre-labels for retraining.
- First version was an offline handheld Raspberry Pi 5 device; moved to the phone app after field tests showed the kit was awkward to carry and overheated in a backpack.
- One unified YOLO11s model replaced the two detectors on the Pi: response time **499 → 351 ms (−30%)**, no loss of accuracy.
- Screened INT8 on four backbones: 2.9–4.2× faster, but **it pushed the two smallest-target pest categories below a usable level on every backbone**; weight-only quantisation kept 96–99% of FP32 accuracy at 24–40% lower energy per image.
- Traced monitoring gaps to the 82 °C software thermal cut-off, not frequency throttling (which cost only 2.6% latency), and removed them in lab tests with a 30 s duty-cycle sleep.
- Ran the two detectors in turn, not at once, avoiding overlaps that slowed the leaf model 58% and the pest model ~150%.
- Benchmarked a Jetson Orin Nano Super (pest detection 186 → 35 ms, at 2.9× the cost and ~1.4× the power) and measured that idle draw dominates the Pi (6.3 W idle vs 8.4 W inferring).

AI Development Intern | Urban Farm Tech, Malaysia *Jan 2026 – May 2026*

Led the firm's first AI project as it expanded from IoT into AI.

- Trained a MobileNetV2 basil disease classifier on ~1,700 images taken on site and exported it to ONNX.
- Deployed it as an offline monitoring node on a Raspberry Pi 5 in the commercial greenhouse, sending a Telegram alert only when 3 of 5 consecutive frames agree, in under 2.5 s.

PUBLICATIONS & OPEN DATA

Lin, D. S. (2026). Thermal cut-off, not frequency throttling: duty-cycle scheduling and a dataset-integrity audit for on-farm edge disease detection in durian. *Smart Agricultural Technology* 15, 102526. doi:10.1016/j.atech.2026.102526. Sole author.

Three further manuscripts submitted: unified vs multi-model edge pipelines, release scheduling for two detectors on one board, and container overhead on edge nodes.

Datasets and measurement code are public on Zenodo (with DOIs) and GitHub; links on the portfolio.

EDUCATION

Bachelor of Computer Science in Data Science with Honours | UCSI University *Sep 2024 – May 2027 (expected)*

Languages: English (fluent), Mandarin (native)